

GIOIA DEL COLLE, Italy

WMO: 163120

Lat: 40.766N

Lon: 16.935E

Elev: 352

StdP: 97.17

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -1.4	(c) -0.1	(d) -6.0	(e) 2.4	(f) 2.8	(g) -4.1	(h) 2.8	(i) 3.1	(j) 16.5	(k) 10.4	(l) 14.1	(m) 9.2	(n) 3.2	(o) 320	(p) 0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.8	(c) 34.9	(d) 21.3	(e) 33.1	(f) 20.7	(g) 31.5	(h) 20.4	(i) 23.7	(j) 29.0	(k) 22.8	(l) 28.1	(m) 22.0	(n) 27.4	(o) 5.0	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	17.7	25.1	21.2	16.6	24.6	20.3	15.6	24.0	72.7	29.1	69.2	28.4	66.2	27.4	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.7	11.6	9.9	-4.5	38.3	1.6	2.1	-5.7	39.8	-6.6	41.0	-7.5	42.1	-8.6	43.6
			-5.0	25.6	1.4	1.1	-6.0	26.4	-6.8	27.0	-7.6	27.7	-8.6	28.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.9	6.4	6.7	9.2	12.4	17.1	21.9	24.6	24.6	20.0	15.9	11.5	7.6
	DBStd	7.14	2.85	2.98	2.88	2.68	2.96	3.25	2.55	2.43	2.96	2.94	3.18	2.98
	HDD10.0	377	115	96	49	9	0	0	0	0	0	1	21	86
	HDD18.3	1861	370	325	283	177	60	7	0	0	16	86	204	333
	CDD10.0	2157	3	4	24	82	219	357	453	452	301	183	67	11
	CDD18.3	599	0	0	0	1	21	114	194	194	66	9	0	0
	CDH23.3	5688	0	0	1	11	213	1113	2023	1873	424	31	0	0
	CDH26.7	2348	0	0	0	0	44	437	916	831	118	1	0	0
(14) Wind	WSAvg	4.2	4.3	4.8	4.8	4.5	3.9	3.8	4.0	3.6	3.7	3.8	4.4	4.2
(15) Precipitation	PrecAvg	672	67	70	71	44	48	40	29	43	51	71	68	70
	PrecMax	988	144	216	173	107	152	189	113	144	149	176	160	120
	PrecMin	322	11	9	9	6	5	2	0	0	1	1	14	30
	PrecStd	218	41	48	42	27	39	42	30	42	43	48	47	26
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.5	16.5	20.2	25.0	30.2	35.2	37.8	37.2	32.9	26.1	20.9	16.1
		MCWB	11.3	10.8	12.3	15.4	19.0	21.5	21.2	21.4	19.6	18.1	15.4	12.8
	2%	DB	13.2	14.3	18.0	21.2	27.8	33.0	35.0	34.8	29.9	24.0	19.0	14.5
		MCWB	10.5	9.9	11.8	14.0	17.7	20.5	21.4	21.2	19.4	17.5	15.4	12.3
	5%	DB	12.1	13.0	16.1	19.7	25.8	31.1	33.1	32.9	27.8	22.2	17.2	13.6
		MCWB	10.0	9.6	11.1	13.3	16.8	20.0	20.5	21.0	19.1	17.0	14.7	11.7
	10%	DB	11.0	11.8	14.2	17.9	23.9	29.1	31.2	31.1	25.8	20.9	16.1	12.4
		MCWB	9.2	9.3	10.4	12.7	16.3	19.4	20.2	20.5	18.6	17.0	14.1	10.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	12.6	14.4	16.9	20.5	24.0	24.9	25.1	23.3	21.2	17.7	14.3
		MCDB	14.3	14.4	17.7	21.3	26.9	30.6	31.4	31.6	26.0	23.4	18.8	15.4
	2%	WB	11.6	11.5	13.0	15.3	19.2	22.4	23.5	23.8	22.0	19.7	16.5	13.1
		MCDB	12.7	13.2	15.8	19.3	25.0	29.4	29.4	28.8	25.3	21.4	17.6	14.1
	5%	WB	10.5	10.6	12.0	14.2	18.2	21.5	22.6	22.9	21.1	18.7	15.4	12.1
		MCDB	11.8	12.2	14.5	17.7	23.4	27.9	28.5	27.7	24.5	20.6	16.8	13.2
	10%	WB	9.5	9.7	11.2	13.4	17.3	20.4	21.7	22.1	20.2	17.7	14.5	11.1
		MCDB	10.8	11.3	13.6	16.4	21.9	26.5	27.9	27.0	23.3	19.8	15.9	12.3
(35) Mean Daily Temperature Range	MDBR		7.0	7.6	8.4	9.4	10.7	11.5	11.8	11.7	9.7	8.6	7.1	6.7
	5% DB	MCDBR	8.5	9.7	11.8	12.7	14.2	14.4	14.9	14.8	12.5	10.8	8.4	7.3
		MCWBR	6.2	6.4	7.0	6.9	6.2	5.5	5.6	6.0	5.4	6.1	5.9	5.6
	5% WB	MCDBR	6.9	7.7	9.5	10.5	12.1	12.6	12.3	12.0	10.2	8.7	6.6	5.9
		MCWBR	5.7	5.8	6.3	6.5	6.3	5.9	5.9	6.0	5.7	5.7	5.3	5.1
(40) Clear-Sky Solar Irradiance	taub		0.330	0.352	0.389	0.437	0.432	0.419	0.414	0.422	0.414	0.383	0.355	0.333
	taud		2.461	2.377	2.275	2.138	2.216	2.293	2.313	2.296	2.312	2.386	2.418	2.453
	Ebn at Noon		824	854	857	836	849	858	859	842	822	807	782	787
	Edh at Noon		78	98	121	148	141	130	127	125	116	96	81	74
(44) All-Sky Solar Radiation	RadAvg		1.85	2.61	3.90	5.12	6.25	7.05	7.22	6.35	4.64	3.24	2.10	1.65
	RadStd		0.19	0.23	0.40	0.40	0.37	0.36	0.30	0.37	0.27	0.26	0.17	0.20

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	-92	N/A	N/A
(47) Regional (11 neighbors)		+0.35	N/A	N/A	N/A	N/A	N/A	-67	+92	N/A

Nomenclature: See separate page